

Graded Lab Tasks – Python Basics for AI

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Allotted Time: 1.5 Hours

Instructions

- Attempt all lab tasks individually.
- Use Python IDLE or any Python environment to implement the tasks.
- Submit your solutions on the CUI portal and provide a GitHub link for your code.

Lab Task 1: Student Grade Calculator

Write a Python program to accept marks of a student (1-100) and display the grade according to the following criteria:

- Grade F if marks are less than 50
- Grade E if marks are between 50 to 60
- Grade D if marks are between 61 to 70
- Grade C if marks are between 71 to 80
- Grade B if marks are between 81 to 90
- Grade A if marks are between 91 to 100

Lab Task 2: Factorial Calculator

Write a program that takes a number from the user and calculates the factorial of that number.

Lab Task 3: Fibonacci Series

Fibonacci series is defined as a sequence where the next number is formed by adding the previous two numbers. Start from 0 and 1. Example:

0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, ...

Steps:

1. Take an input number that shows how many terms to display.
2. Use loops to display the Fibonacci series up to that term.
3. Example: If the input number is 6, the output should be: 0, 1, 1, 2, 3, 5.